

WASP-43b

R-1602

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_220122 · created 2026-08-02T02:25:20+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459601.9311 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.023
Transit depth (Rp/Rs)^2	2.87 +/- 0.78 [%]
Orbital Inclination (inc)	81.33 +/- 1.76
Airmass coefficient 1 (a1)	0.962 +/- 0.016
Airmass coefficient 2 (a2)	0.026 +/- 0.03
Scatter in the residuals of the lightcurve fit is	1.63 %
Best Comparison Star	None
Optimal Aperture	3.56
Optimal Annulus	11.48
Transit Duration (day)	0.0483 +/- 0.0072

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.169 ± 0.023 vs 0.1594 (deviation 0.42σ)
Duration fit vs published	0.0483 d vs 0.0512 d (deviation 0.41σ)
Mid-time O–C	-3.2 min
Depth SNR	7.3
Residual scatter	1.63 %

Light curve

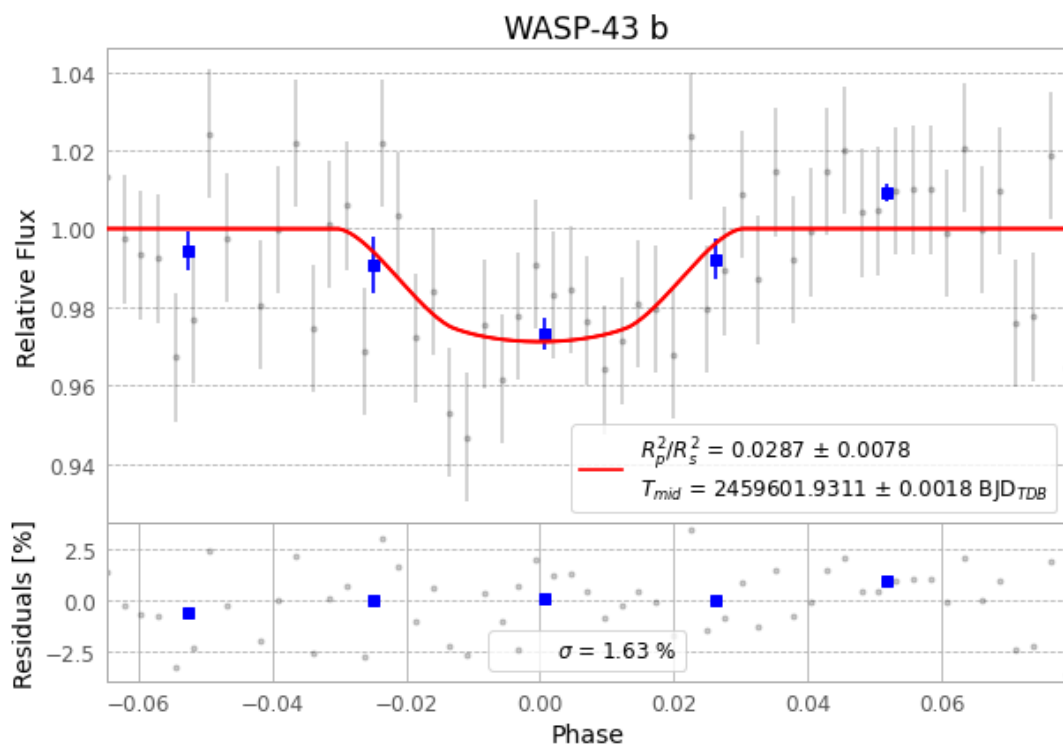


Figure 1 — FinalLightCurve_WASP-43 b_2022-01-22.png (SHA-256 67bbccee47e2e154...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 262], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c45797547c4fdcd9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

57 FITS frames (37 MB), WASP-43220122085713.FITS → WASP-43220122114513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-220122.log (be3f866af373...), FinalLightCurve_WASP-43 b_2022-01-22.png (67bbccee47e2...), FinalLightCurve_WASP-43 b_2022-01-22.csv (f86676e812c6...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 02:25Z.